

STA561-Probabilistic Machine Learning

Final Project

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1 Overview

In the realm of e-commerce, discerning the causal effect of advertising on purchase decisions is crucial. Our project aims to employ uplift modeling techniques within the Taobao E-commerce environment to isolate this effect. Unlike traditional predictive models that focus on direct outcome prediction, uplift models are designed to forecast the incremental value or *lift* that results from a specific treatment. This is formally expressed as the difference in the probability of a purchase given the presence of treatment versus its absence:

$$\text{Lift} = P(\text{buy} \mid \text{treatment}) - P(\text{buy} \mid \text{no treatment}) \quad (1)$$

In contrast, conventional models typically predict the outcome directly as a function of treatment:

$$\text{Outcome} = P(\text{buy} \mid \text{treatment}) \quad (2)$$

Our objective is to tailor and evaluate various probabilistic machine learning models, such as causal forest algorithms, XGBoost, and SVM, to predict uplift. These models will be adapted to the high-dimensional data characteristic of the Taobao user base, incorporating elements like user activity, historical transactions, and demographic details to mitigate confounding influences and accurately estimate the effect of advertisement exposure.

A comprehensive cross-validation strategy will be employed to assess model performance, complemented by the implementation of propensity score matching to refine control group selection, thereby strengthening the causal interpretations of our analysis. The challenge lies in specifying the propensity model and adjusting the treatment effect model accordingly to accommodate the sensitivity of this specification. With Taobao's extensive datasets, we aim to address these challenges and contribute meaningful insights into the effectiveness of advertisement interventions in e-commerce.

2 Dataset and EDA

3 Deliverables

We describe our approach to applying uplift modeling, which is central to our deliverable. This includes mathematical formulations and statistical techniques. The comprehensive report is structured as follows:

3.1 Methodology

3.1.1 Data Preparation

We merge user-level data from Taobao's databases to form a comprehensive dataset. For each user, represented by a unique User ID, we track the following variables:

- X_i - A vector of user characteristics and historical data.
- T_i - A binary treatment indicator where $T_i = 1$ if the user clicked on an advertisement, and $T_i = 0$ otherwise.
- Y_i - The outcome variable, such as purchase amount or engagement level, post-treatment.

3.1.2 Model Specification

We specify the uplift model as follows:

$$Y_i = \mu(X_i) + \tau(X_i) \cdot T_i + \epsilon_i$$
$$\tau(X_i) = E[Y_i|X_i, T_i = 1] - E[Y_i|X_i, T_i = 0]$$

Here, $\mu(X_i)$ represents the expected outcome without treatment, $\tau(X_i)$ is the conditional average treatment effect (CATE), and ϵ_i is the error term.

3.1.3 Estimation Techniques

In our project, since our data is balanced, we employed the famous Two-Model approach (T-learner) CATE algorithms.

- **Two-Model Approach:** We estimate $\mu(X_i)$ and $\tau(X_i)$ separately using two distinct models.

$$\hat{\mu}(X_i) = f_1(X_i; \theta_1)$$
$$\hat{\tau}(X_i) = f_2(X_i; \theta_2) - f_1(X_i; \theta_1)$$

where f_1 and f_2 are the model functions with parameters θ_1 and θ_2 , respectively.

We then explore and incorporate a variety of Machine Learning algorithms, such as Random Forests and Neural Networks, into the famous Two-Model approach (T-learner), to estimate $\tau(X_i)$:

- *XGBoost T-learner:* This method fits two gradient boosting models $\hat{\mu}_1$ and $\hat{\mu}_0$ using data from treated and control units respectively. The fitted CATE function is $\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$. All parameters were set at their default values in XGBClassifier function.
- *Random Forest T-learner:* Similar to *XGBoost Forest T-learner*, we used random forest models to fit the treated and control units to estimate $\hat{\tau}(x)$. All parameters were set at their default values in RandomForestClassifier function.
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3.1.4 Evaluation Metrics

We use several metrics to evaluate the performance of our uplift models:

- Uplift Curve
- Qini Coefficient
- Incremental Revenue

Each of these metrics provides insight into the effectiveness of the advertising campaign, allowing for an informed decision-making process.

3.2 Results and Discussion

Within the deliverable, we will present the results of the uplift models, including estimated CATEs and their impact on the Taobao platform. Our discussion will include strategic insights derived from the analysis and potential recommendations for future campaigns.

3.3 Conclusion

We conclude our deliverable by summarizing the application of uplift modeling to Taobao data, emphasizing the enhancement of marketing strategies and driving meaningful engagement on the platform.

Table 1: Optimal Weights for Different Models

Model	Optimal Weight
Logistic Regression	0.25001355
XGBoost	0.24999748
Random Forest	0.24999235
Neural Network	0.24999662

Table 2: Qini Index and IRR for Different Models

Model	Qini Index	IRR
Logistic Regression	0.05324	0.1108
XGBoost	0.06702	-0.0196
Random Forest	-0.02864	0.4588
Neural Network	-0.04815	-0.0049
Ensemble Model	0.5843	0.4251